

nationalgrid



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Title of Work - *To what extent does National Grid's organisational strategy, management practices, and governance framework reflect genuine sustainability? A theory-informed evaluation of the tools, structures, and strategic choices underpinning its progress and limitations.*

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N.B. Figures 1-7 are present in the appendix + focuses on National Grid only in its UK context.

1. Introduction

The Friedman doctrine implies that the sole social responsibility of a firm is to maximise shareholder value within the rules of the market, implying sustainability objectives are only legitimate if they enhance profitability or competitive advantage (Friedman, 1970). This perspective remains a 'critical touchstone' in debates surrounding the role of business in society, and continues to influence contemporary sustainability management by framing sustainability as a means of efficiency and value creation rather than an ethical or ecological obligation (Jain, 2024). Richard Branson echoed this perspective, stating that the only way to stop climate change is for industry to profit from it" (Branson, 2012). This essay creates its own criteria for sustainability, rejecting purely profit-aligned conceptions in favour of a systemic perspective that evaluates an organisation's practices, strategies and governance against a theory-driven understanding of sustainability, irrespective of alignment with commercial incentives. National Grid (NG) has been selected as the case study due to its position as a regulated natural monopoly within the UK energy system, granting the firm significant influence over national decarbonisation pathways. Being a natural monopoly, NG is regulated by law, and provides a unique opportunity to analyse the impacts of a regulator on sustainability objectives.

2. Criteria for "Genuine Sustainability"

Systemic sustainability represents a departure from linear or siloed thinking, recognising the interconnectedness of environmental, social and economic systems, focusing on addressing the root causes rather than fixing isolated issues (SD, 2025). When analysing NG's policies, systemic sustainability is used as a foundational

benchmark for judging decarbonisation. Dittmar's (2014) 'Negation Concept' argues that sustainability should be defined by removing the system-level drivers of unsustainability such as: fossil fuel dependence, infrastructural lock-in and the inefficient use of renewables rather than promoting vague ecological ideas. At the organisational level, scholars stress that genuine sustainability involves deep, long-term business transformation, with Banerjee (2002) arguing that environmental commitments only become meaningful when they are embedded into operational routines. Del Baldo and Baldarelli (2017) extend this by stating sustainable business models require firms to 'transform their business logic', while re-orientating their value-creation process. Finally, stakeholder theorists argue that true sustainability requires integrating stakeholder interests into business decision-making. There is also an acknowledgement that organisations operate within complex social and regulatory systems which help shape responsible business practices (Horisch et al., 2014; Del Baldo, 2017). Drawing these perspectives together, this essay defines genuine sustainability as practices that contribute to systemic decarbonisation, embeds sustainable objectives into long-term business transformation, efficiently engages with stakeholders to improve sustainability and directly addresses the most material and structurally embedded sources of harm.

3. How National Grid is Regulated

NG exists as a monopoly in the UK energy market and is therefore regulated by Ofgem, an independent energy regulator, under the RIIO price control network. This framework governs revenues, investment allowances and performance obligations by imposing price controls and service standards while guiding the long-term

investment pathway for decarbonisation. Consequently, the firm's decisions are partially constrained by regulatory requirements, an essential piece of context for certain sustainability initiatives discussed throughout this essay (National Grid, 2025a).

4. Innovation and National Grid

Innovation is essential for businesses to be able to address challenges such as climate change and resource depletion, enabling the development of new technologies, practices and solutions which improve operational sustainability (Visnjic, 2023). Bill Gates (2021) perspective adds that long-term sustainability is unimaginable without strong innovation.

4.1 Grid Transformation

NG has committed to 17 projects which enable the use of offshore wind, not by generating renewable power itself, but by creating conditions that allow wind farms to operate at scale. The company's Great Grid Upgrade and Accelerated Strategic Transmission Investment programs represents the largest transformations of the UK transmission network in 50 years and aims to connect up to 50 GW of offshore wind power by 2030 (National Grid, 2025c). A coordinated set of investments followed, including new long-distance subsea HVDC links, reinforced onshore corridors and upgraded substations, all designed to transport renewable electricity from the Dogger Bank windfarm to population centres where energy is required. These policies clearly match the criteria of a systemic business change defined in Section 2, actively reconfiguring the basic architecture of the UK power system so that large

volumes of offshore wind can replace coal and gas. Dependence on fossil fuels and the continued usage of infrastructure that is unable to facilitate renewables can be categorized as unsustainable using Dittmar's (2014) criteria. Using this logic, the proposed pipeline investment of 14.82 billion (RIIO-T3, 2024) aims to expand and extend the network for 'new clean sources of power', showing concrete green intervention to reduce fossil fuel dependency. Therefore, by increasing grid capacity, NG will reduce the renewable energy wasted through the phenomena known as curtailment, addressing a core structural barrier to sustainable energy, while also reducing national dependence on fossil fuels.

4.2 Elimination of Sulfuric Hexafluoride

Another innovative policy deployed by NG is the elimination of Sulfuric Hexafluoride (SF_6), an extremely high-impact greenhouse gas widely used in electricity transmission and distribution to insulate live electrical parts and safely switch the flow of electrical current on and off (National Grid, 2025d). NG has already reduced SF_6 by 33%, with Ofgem awarding £8.5 million to develop a long-term strategy to reduce SF_6 dependency (National Grid, 2025c; Ofgem, 2024). A case study in 'Annual Environmental Report' details the construction of the first SF_6 -free transmission substation using an alternate insulating gas called EconIQ. All high voltage insulating gas leak; however, EconIQ has a global warming potential (GWP) of 298, compared to 23,500 for SF_6 (National Grid, 2025c). Consequently, this reduction and proposed elimination represents a significant intervention, requiring NG to alter the core of its transmission infrastructure rather than greening a peripheral process. Given that SF_6 is deeply embedded within the engineering logic

of high voltage systems, its removal necessitates substantial transformation, aligning with Banerjee's (2002) conception of corporate environmentalism, whereby environmental objectives are integrated into core operational practices rather than treated as ancillary initiatives.

However, despite these advances, SF₆ still remains responsible for around 90% of its Scope 1 and 2 emissions (National Grid, 2025c), and the slow pace of elimination suggests the company is still far away from addressing its most material environmental harms. The partial progress raises concerns on whether the company can move quickly enough to eliminate SF₆. Although progression towards SF₆-free substations may appear unimpressive and un-sustainable, it reflects the reality of meaningful sustainability development over time. Schendler (2009) argues that large organisations achieve environmental transformation not through rapid, perfect shifts but through gradual path dependent progress, constrained by technical, financial and cultural criteria. The headway made in targeting a process which is deeply embedded into the engineering logic, while reducing the reliance on a high impact pollutant, aligns with genuine sustainability. The literature identifies this as essential to authentic organisational sustainability, with Schendler (2009) stating that if sustainability was straightforward, everyone would be doing it.

4.3 Interim Conclusion

On balance, NG's innovation strategies reflect structural intervention instead of efficiency-led optimisation, directly enabling renewable-energy integration and fossil fuel displacement. While SF₆ remains a major source of emissions, Schendler's

(2009) perspective on path dependence explains the slow progress and discredits the idea of slow pace equating to unsustainable practices.

5. National Grid's Stakeholder Engagement

Horisch et al. (2014) argues sustainability cannot be delivered solely through internal corporate action, and instead emphasises the firm's ability to integrate both environmental and social objectives across its network as key factors in achieving genuine sustainability. This perspective requires aligning stakeholder interests with the long-term trajectory of the business.

5.1 Suppliers

NG has committed to implementing science-based targets (SBT's) across the top 80% of suppliers responsible for the largest share of emissions, with 57% implementing such targets in 2025 compared to none in 2024 (National Grid, 2025c). SBT's align with what climate science necessitates to limit global warming, reinforcing their legitimacy as a governance mechanism across value chains (SBT, 2025). The choice to pursue SBT's rather than net-zero is a strong starting point for sustainability, as net-zero is often criticised for focusing more on balancing emissions rather than cutting polluting activities, still allowing for dangerous warming (Bevan, 2021; Steiner et al., 2021). Instead, SBT's set targets compatible with less than 2 degrees of warming, emphasising cutting emissions rather than balancing them (Piper, 2021). NG's commitment exemplifies stakeholder engagement to improve not only its own procurement sustainability, but also that of its most polluting suppliers. By building a supply-chain network conscious of its emissions, NG shows their ability

to embed environmental objectives into long-term business transformation by working with suppliers to implement more sustainable practices.

However, the choice to pursue a policy of encouragement in the context of carbon reduction targets (CRT's) may cause concern, as there has been no change in the percentage of suppliers who have committed to said targets between 2024 and 2025 (67%). Suppliers eligible for these targets comprise the top 75% of firms based on spend rather than emissions. This highlights a potential flaw in NG's governance approach, whereby assertive mechanisms are applied to SBT alignment, but are not extended to the carbon-reduction pathways of all suppliers, despite the scale of Scope 3 emissions associated with procurement activities (Figure 2). This is also significant given the reputation of CRT's for often falling short in climate change mitigation (Hollins, 2025). Where supplier commitments remain voluntary, firms may delay or avoid compliance, especially given the technical, financial and cultural constraints outlined by Schendler (2009). Conversely, behavioural psychologist Brehm (1996) provides an explanation to why NG has chosen this level of coerciveness when dealing with suppliers. When actors perceive their autonomy to be threatened, they are more likely to delay compliance rather than engaging constructively, known as Reactance Theory. Given this, NG's decision to exercise assertion only on the most polluting suppliers reflects a balance between maintaining operational continuity and advancing sustainability objectives, rather than a failure in stakeholder governance. Furthermore, this essay commends NG for choosing a more influential carbon reduction target for the highest-emitting supplies, instead of an option that would merely deliver performative results.

5.2 Consumers

The company's stakeholder engagement also extends downstream to societal stakeholders. NG has committed to an ongoing support network for those unable to meet energy costs, establishing a £13.8 million “good energy affordability fund”, which has already benefited 259,884 households across the UK. In addition, the company has also logged 25,237 hours of volunteering since 2021, to support local communities (National Grid, 2025a). These actions align with Kate Raworth’s concept of Doughnut Economics (*Figure 3*), which defines sustainability as the ability of an economy to meet society’s basic social foundations, while staying conscious of environmental implications (Raworth, 2017). Using a more micro-economic lens, this branch of NG’s stakeholder engagement directly addresses the societal shortfall of energy affordability. The UK has been experiencing an energy crisis (Parliament, 2025), with household energy debts reaching a total of £4.43 billion in Q2 of 2025 (EFPC, 2025). By targeting the crisis, NG demonstrates engagement with a material shortfall identified within Raworth’s framework. This is reinforced by Ofgem attitudes, which emphasise protecting vulnerable consumers as a core objective of energy governance (Ofgem, 2025).

5.3 Interim Conclusion

On balance, NG’s stakeholder engagement aligns with genuine sustainability through a targeted governance strategy that implements the most stringent emissions standards to the highest emitting suppliers, thereby addressing the most material sources of harm. Consumer incentives follow a similar narrative, focusing on

a material shortfall within the UK economy. This reflects the same logic as supplier engagements, demonstrating alignment of social and environmental objectives across stakeholder governance.

6. Environmental Policies

Stakeholders are not defined by their ability to speak or contract, but whether they are affected by corporate activity and essential to organisational value creation. Not only does nature meet this definition, but by treating nature as a resource, one's attitude will reflect one of a narrow and anthropocentric reading of stakeholder theory (Starik, 1995).

6.1 Biodiversity Net Gain Targets

The UK is one of the most nature-depleted countries in the world, with a score of 53% on the 'biodiversity intactness index' (National Grid, 2025c). As of February 2024, a mandatory 'Biodiversity Net Gain' (BNG) target was introduced by the UK government, requiring developers to deliver a baseline of a 10% net increase in biodiversity on major projects (England, 2024). *Figure 4* outlines the legal requirements for BNG regulations, prohibiting companies from immediately skipping straight to "offsetting" without demonstrating reasonable consideration of the above policies (IntegratedLM, 2024). NG's disclosures indicate a high level of compliance and engagement with the BNG framework, reflecting organisational alignment with national biodiversity objectives. BNG targets are considered by the company in the very early stages of development, with avoidance prioritised where habitats are important (National Grid, 2025c). Where construction cannot be avoided, NG uses

the DEFRA calculator along with other natural capital tools to quantify impacts. In situations where BNG is not mandatory, NG applies these steps to the Town and Country Planning Act 1990, highlighting the firm's desire to go 'above and beyond' with environmental improvements. As per *Figure 5*, NG has consistently exceeded the 10% minimum requirement and aims for all their projects to meet a 15% net gain in value (National Grid, 2025c).

However, Ofgem has recently blocked NG's proposal to normalise exceeding the 10% government requirement, leading the regulator to face criticism for hindering nature recovery. While there are certainly grounds for such criticism, one must remember the definition of systemic sustainability outlined in Section 2. Ofgem's reluctance to approve the proposal made by the company is based on consumer affordability, as the cost burden for this proposal would be passed down to consumers who, as already established, are facing an energy crisis (Pashby, 2025; Consultation, 2025). As such, while this essay commends NG's environmental ambition, the regulatory constraint imposed by Ofgem reinforces the need to evaluate sustainability not by the maximisation of a single target, but by the resilience and coherence of the system as a whole. On balance, NG's consistent compliance and engagement indicates that the firm's environmental policies reflect genuine sustainability within the structural limitations of a regulated energy system.

6.2 Waste Management

The global construction industry makes up $\frac{1}{3}$ of the world's overall waste (National Grid, 2025c), which necessitates efficient waste management strategies in

construction. Given NG's role in transforming the energy sector in the UK to be more sustainable, construction activities are unavoidable and therefore must be managed correctly in order to match sustainability objectives. Landfill sites are renowned for causing significant environmental harm, while simultaneously reducing societal and economic welfare (Matthews, 2023). NG's decision to implement a Life Cycle Assessment (LCA) ensures that an effective waste management strategy is in place, by tracking the total environmental impact of a product, process or service from start to finish (Quist, 2024). As illustrated in *Figure 6*, the 'closed loop' model attempts to turn waste into inputs for new products, which successfully diverts waste from landfill sites. The company has disclosed that 84.6% of construction waste was recycled, with a further 14.8% reused (National Grid, 2025c). These figures are undoubtedly strong, demonstrating the firm's elimination of landfill use in its construction activities and its engagement with one of the most environmentally damaging aspects of waste management. More broadly, NG has exceeded recycling goals in their substations, reaching a 88.5% recycling rate compared to a 60% benchmark, while also hitting a 54% recycling rate across offices with a waste reduction stat of 33% from a 2020 benchmark (National Grid, 2025c). While not as impressive as the construction statistics, these stats represent the attitudes of the company to improve resource circularity, placing particular emphasis on the sector which contributes the most to waste.

6.3 Interim Conclusion

On balance, the evidence suggests that NG has demonstrated a genuine commitment to environmental policy, frequently exceeding mandatory targets where

feasible. The regulator-firm interaction illustrates how Ofgem moderates ambition to promote systemic coherence rather than linear target maximisation. NG's waste management strategy aligns with Dittmar's (2014) negation concept, removing a system-level driver of unsustainability within construction, while simultaneously improving waste management strategies across their operations. Taken together, the environmental policies align with the criteria established in Section 2 by directly addressing some of the UK's most material forms of ecological harm: biodiversity loss and the environmental impacts of landfill.

7. Financial Health of National Grid

At this point, it's clear that NG is operating within the grounds of genuine sustainability, but are they doing so in an economically viable way? Del Baldo & Baldarelli (2017) argue that sustainability cannot be achieved by rejecting economic logic, and for sustainability to exist, a firm must remain economically viable. Horisch et al. (2014) also emphasises the importance of including economic stakeholders into the process of aligning interests. *Figure 7* illustrates financial metrics which align with economic viability, with Buffet's (2007) conditions being used as benchmarks due to his focus on long-term resilience instead of short term profit-maximisation. Despite strong sustainability initiatives, NG still maintains the financial characteristics which are attributable to a firm with strong financial health, reinforcing the compatibility of regulatory oversight with sustainable business practice.

8. Conclusion

Taken together, this essay has evaluated NG against a theory-driven criteria of genuine sustainability which departs from western, profit aligned approaches outlined in the introduction. The reader is prompted to revisit section specific conclusions, which, despite engaging in evaluations, consistently come to the conclusion that NG successfully addresses the structurally embedded sources of environmental and social harm rather than relying on symbolic or efficiency-based interventions. The analysis demonstrates that regulatory oversight has strengthened NG's sustainability outcomes by embedding societal and environmental objectives within decision making, while also remaining attentive to economic viability of the firm. Therefore, NG can be credibly characterised as a firm which is operating within the grounds of genuine sustainability.

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10. Appendix

Figure 1

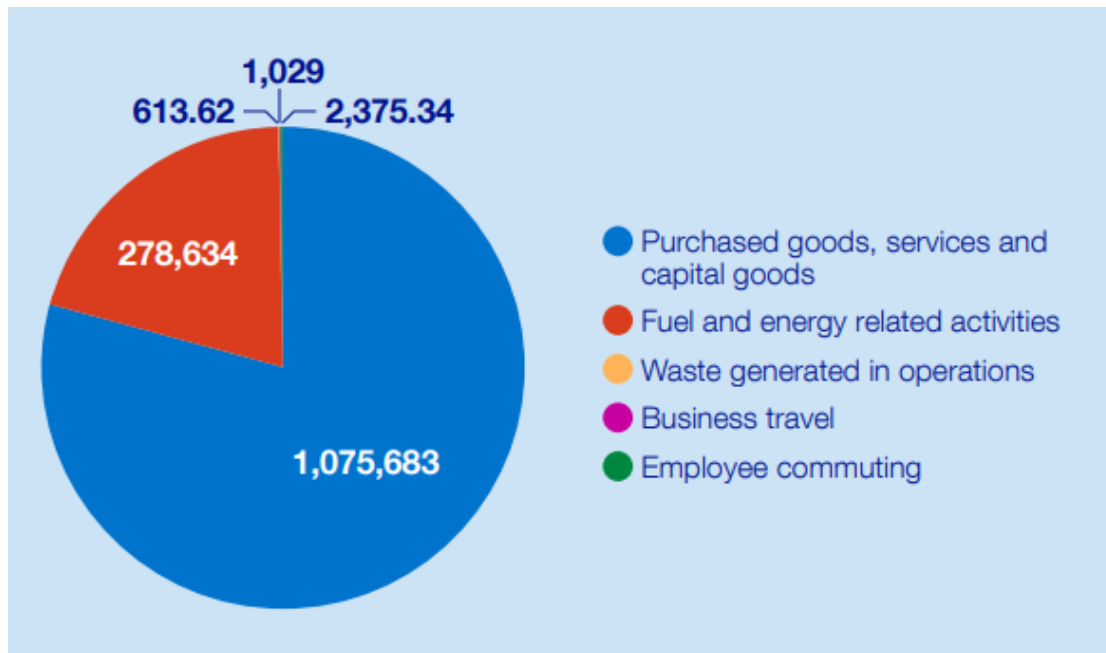
Business & the Natural Environment (MN20448) Assessment Questions 2025

Below you have an option of six essay questions. Please answer ONE questions. The total essay length is 3000 words. The usual 10% leeway for word length applies, which excludes references. Please attend the seminars where we'll go more in-depth on how to best approach the essay.

1. Using philosophical ideas and/or ethical theories concerning the human–nature relationship, analyse a real-world example in which organizational activity has led to significant environmental consequences (positive or negative). In your answer, reflect on how ethical and/or philosophical theory can be used to improve the organization's approach.
2. Is “green capitalism” a genuine path to sustainability or just a convenient illusion? Take a clear stance and justify your argument using theoretical insights and evidence.
3. Identify an organization you believe is close to achieving genuine sustainability. Explain how it did so, focusing on the organizational tools, management practices, and strategic choices that made this possible. In addition to these practical measures, your answer should also be informed by theory and concepts to assess why the organization's approach works (or where it may still fall short).
4. Researchers suggest that the behaviour of the richest within society- those of high socio-economic status- must be changed to avert climate catastrophe. Do you agree? Explain why/why not by focussing on one behaviour that could be changed.
5. Organisations can encourage or discourage climate action through the use of mass communications. Do you agree? Explain why by choosing one organisation and critically assessing their climate communications.
6. Human dominion over other animals is essential for a flourishing society. Do you agree? Explain why by focussing in on one relationship between humans and non-human animals.

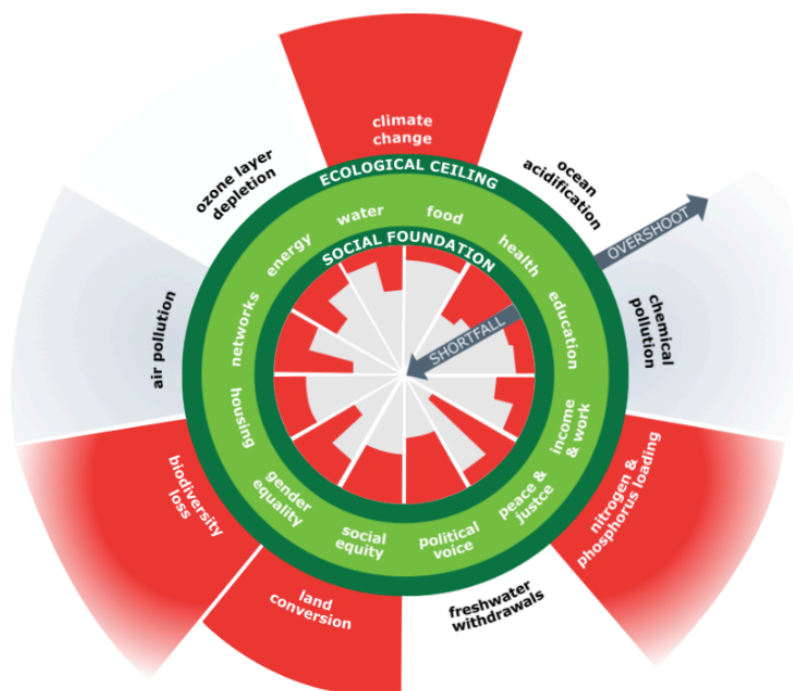
(assignment brief, question 3 selected)

Figure 2



(National Grid, 2025c)

Figure 3



(Raworth 2017)

Figure 4



(IntegratedLM, 2024)

Figure 5

Project name	Baseline units	Committed % net change
Bradford West	1.32	15.00
Cellarhead	1.18	10.00
Cilfynydd	9.08	16.88
Creyke Beck Birkhill Wood	23.71	10.00
Creyke Beck Wanlass Beck	54.21	10.00
Didcot	7.10	15.00
Elstree B (Letchmore Heath)	94.40	15.00
Ferrybridge B SSE BESS	2.68	15.00
Fleet	4.89	15.00
Grain to Tilbury (TKRE)	80.09	15.10
Laleham	0.95	15.00
Margam	157.40	10.00
Melksham CC Grid Park	0.25	28.00
North London reinforcements (HWUP)	328.91	10.00
Pitsmoor	4.64	10.00
Rainhill	-	10.00
Ryehouse	19.21	15.00
Southshields Smartwires	31.99	15.00
Wymondley	0.38	15.00

(National Grid, 2025c)

Figure 6



(Golsteijn, 2022)

Figure 7

Gross Profit Margin	69.00	Greater Than 40%	Good Investment	Total Revenue	19,584.00	N.B. 4 years averages used for accuracy
Return-On-Equity	14.00	Above 10%	Good Investment	COGS	6,035.00	
Debt to Shareholders Equity	2.40	Less Than 0.80	FALSE	Net Earnings	3,881.00	
Net Earnings Margin	20.00	Greater Than 20%	Good Investment	Shareholders Equity	27,770.00	
Dividend Payout Streak	21.00	Greater Than 15yrs	Good Investment	Total Liabilities	67,527.00	
(rounded to 2sf's)						

Brief justification: Collectively, these metrics indicate the economic viability through stable returns, consistent dividends, and resilient margins. Although leverage appears high, elevated debt-to-equity ratios are characteristic of regulated, capital-intensive sectors, where predictable revenues and regulatory oversight support long-term financial stability rather than speculative risk. (Buffet 2008; Maverick, 2025; Yahoo, 2025)